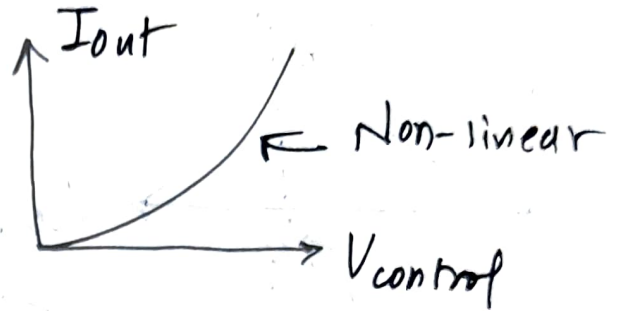
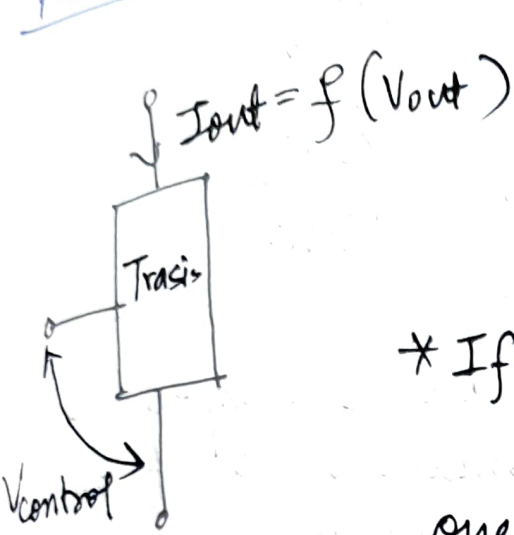


Analog IC Design - Intro to Gm/Id Method

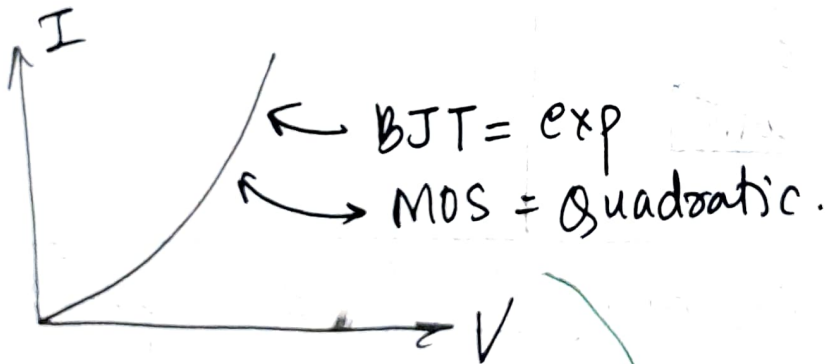
part-1: Transistor Efficiency



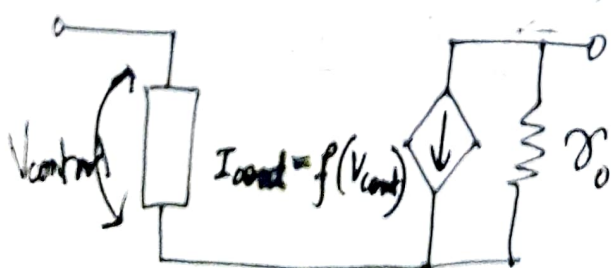
* If Transistor replaced with BJT ;
one would have an exponential-graph

* MOS-Transistor

→ would give quadratic graph

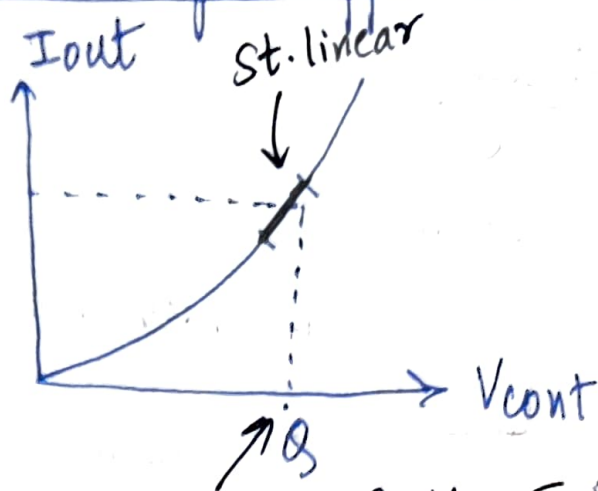


Model this non-linear graph into voltage source



→ non-ideal
has some resistance
 r_o .

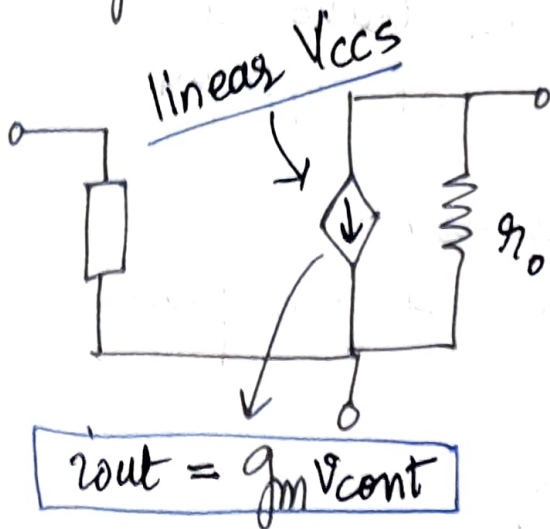
Small Signal Approx: Linear VCCS



$$g_m = \frac{\partial I_{out}}{\partial V_{cont}}$$

$$g_m = \frac{i_{out}}{v_{cont}}$$

Small change in V_{cont} results in small change in I_{out} .



I_Q are we expecting small value of g_m or large value due to this small change in V_{cont} .

Why g_m defines the gain of the amplifiers.

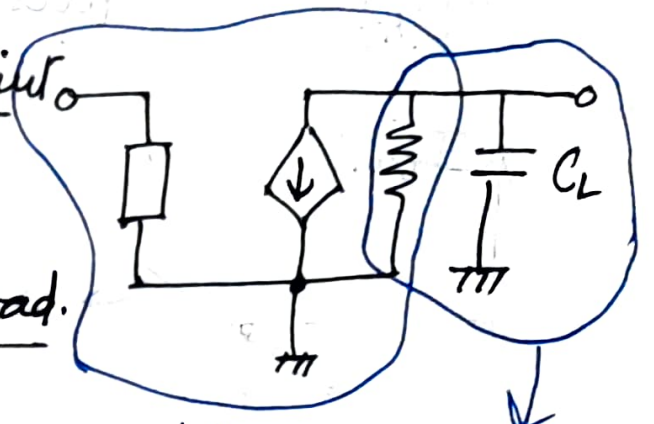
$$A_v = \frac{v_{out}}{v_{in}} = -g_m r_o = -\frac{g_m}{I_Q} \cdot V_A$$

high gain $A_v \rightarrow$ high g_m

② g_m controls the speed.

$$\tau = r_o C_L \Rightarrow B.W = \frac{1}{2\pi\tau}$$

represents the time constant of the circuit.



* to define time-const. we need capacitive load.

also defined as $\tau = r_o C_L \Rightarrow B.W = \frac{1}{2\pi\tau}$ { a freq bandwidth } current Src drives RC network.

$$GBW = |A_{v\tau}| \cdot BW = f_u = \frac{g_m}{2\pi C_L}$$

(gain-BW product)

$|A_{v\tau}|$ cas amplifier is going to be attached to -ve feedback

to have high-speed circuit $\Rightarrow$ increase g_m

③ g_m controls the Noise { both MOSFET thermal noise & BJT shot noise }

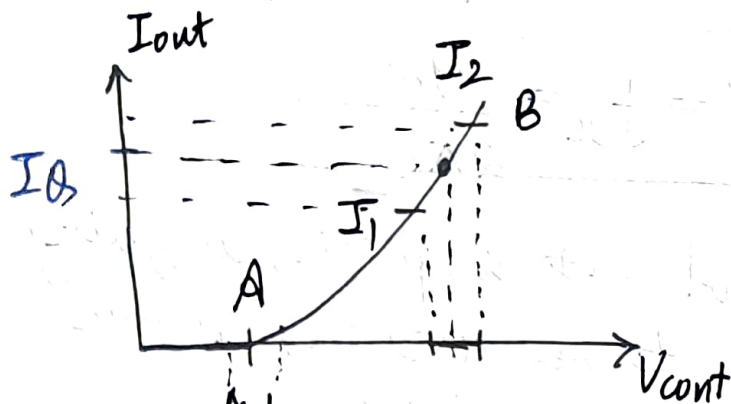
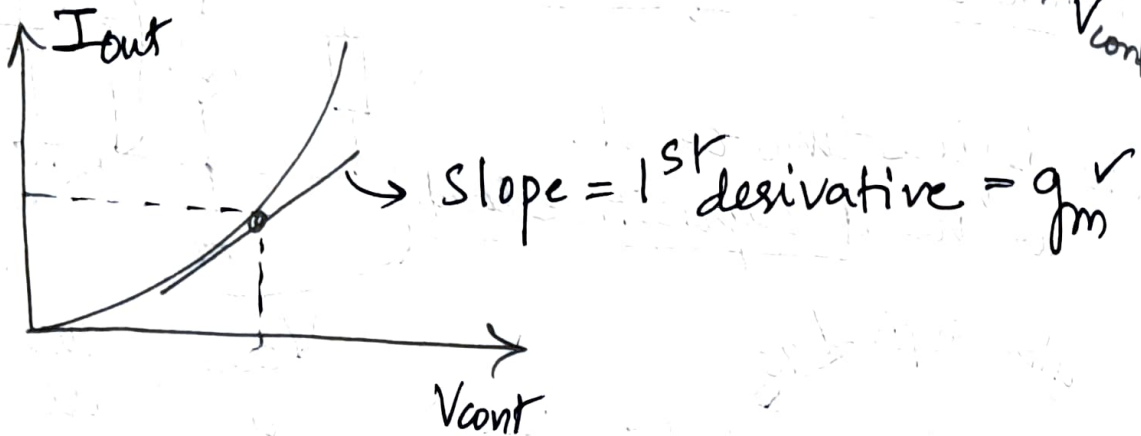
$$\downarrow \mathcal{V}_{n,in}^2(f) \propto \frac{1}{g_m \tau}$$

input-referred noise

we need to keep this as small as possible $\therefore g_m \uparrow \checkmark$

We must pay I_Q to buy g_m

$$g_m = \frac{\partial I_{out}}{\partial V_{cont}} = \text{1st derivative of this graph}$$



(i) at A $\rightarrow g_m = \frac{\partial I_{out}}{\partial V_{cont}} = \frac{I_2 - I_1}{V_{cont}} = \text{Slope of } I_{out} \text{ vs } V_{cont}$

(ii) at B: a small change at B results in large changes in I_{out} (I_2 & I_1)

• to have a very high value of g_m ; I've to pay a large bias current I_Q

* high bias current $\Rightarrow$ high power consumption.

$$I_B \rightarrow \boxed{\text{Transistor}} \rightarrow g_m$$

Ratio of what you want (for g_m) we pay it by supplying ~~the~~ current (I_B).

$$\text{Transistor Efficiency} = \boxed{TE = \frac{\text{What you want}}{\text{What you pay}} = \frac{g_m}{I_B}}$$

BJT - done later - not included here.

Mathematical insight into g_m/I_C

$$\log_b(y) = x \rightarrow b^x = y$$

$$\log_{10}(y) = x \rightarrow y = 10^x$$

$$\ln(y) = \log_e(y) = x$$

$$\checkmark e^x = y$$

$$\bullet \ln(e^x) = \log_e(e^x) = x$$

$$\bullet e^{\ln(x)} = e^{\log_e(x)} = x$$

derivative of $\ln(x)$ w.r.t x

$$\frac{d}{dx} \{ \ln(x) \} = 1/x$$

Replace x with a fⁿ of $y(x)$

$$\rightarrow \frac{d}{dx} \ln(y) = \frac{1}{y} \frac{dy}{dx}$$

$$y = I_c \quad x = V_{BE}$$

$$\Rightarrow \frac{d}{dx}(\ln(y)) = \frac{1}{y} \cdot \frac{dy}{dx}$$

$$\frac{d}{dx} \ln(I_c) = \frac{1}{I_c} \frac{d}{dx} (I_c)$$

$$\frac{d}{dx} \ln(I_c) = \frac{1}{I_c} \left(\frac{dI_c}{dV_{BE}} \right)$$

g_m represents the slope of I_c

$$\therefore \frac{d}{dx} \ln(I_c) = \frac{g_m}{I_c}$$

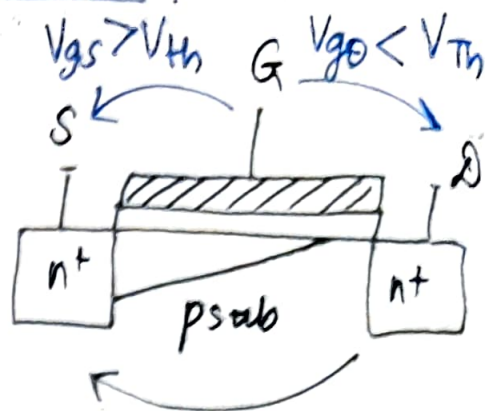
vs

g_m/I_c is the 1st derivative of log of current

The Old School Mosfet model

• N-type channel region
(inversion layer) formed

at: $V_{gs} > V_{th}$



• Threshold voltage: $V_{th} \sim 0.3V - 1V$ $V_{ds} > V_{gt}$

① $V_{gs} > V_T$

② $V_{ds} \gg (V_{gs} - V_T)$

$V_{ds} \geq V_{ov} \rightarrow$ overdrive voltage

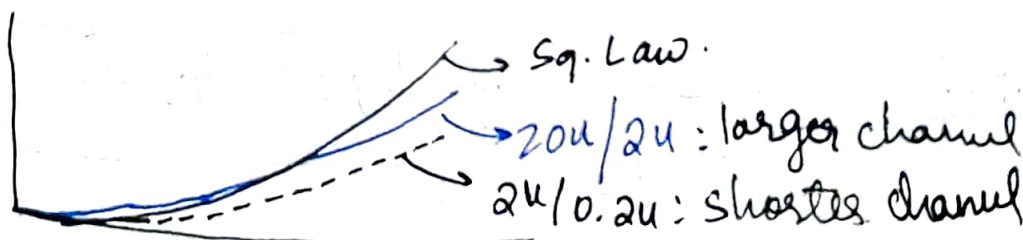
current eqⁿ in the saturation region:

$$I_D = \frac{1}{2} K'_n \left(\frac{W}{L} \right) V_{ov}^2$$

MOSFET IV - characteristics

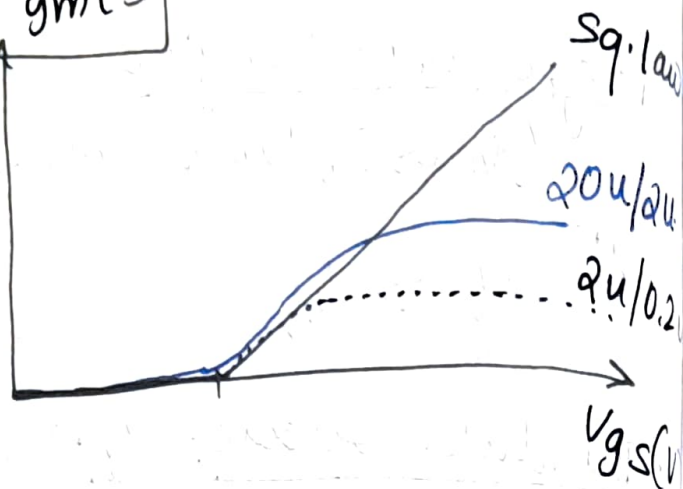
• $I_D = 0$ if $V_{gs} < V_{th}$: cutoff region

• The Square law: $I_D = \frac{1}{2} K'_n \left(\frac{W}{L} \right) V_{ov}^2$



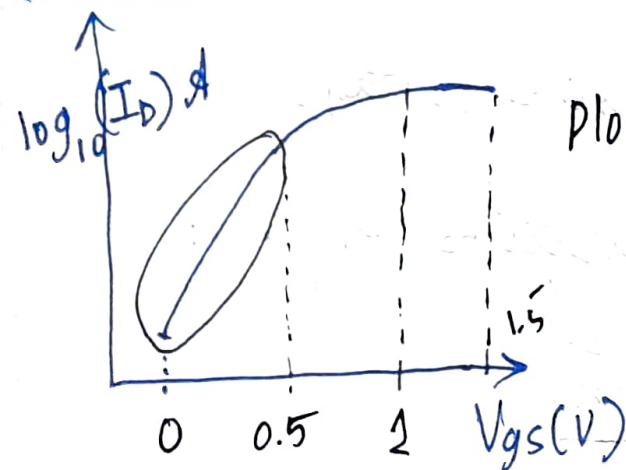
$$g_m = \frac{\partial I_D}{\partial V_{gs}} = k'_n \frac{W}{L} V_{ov} \quad \text{gm(s)}$$

* Why is there a significant deviation?



→ 2nd order laws.

Real device IV characteristics



Plot of o/p character of real MOS.

say $V_{th} = 0.5V$ and we said $I_D = 0$ if $V_{gs} < V_{th}$.

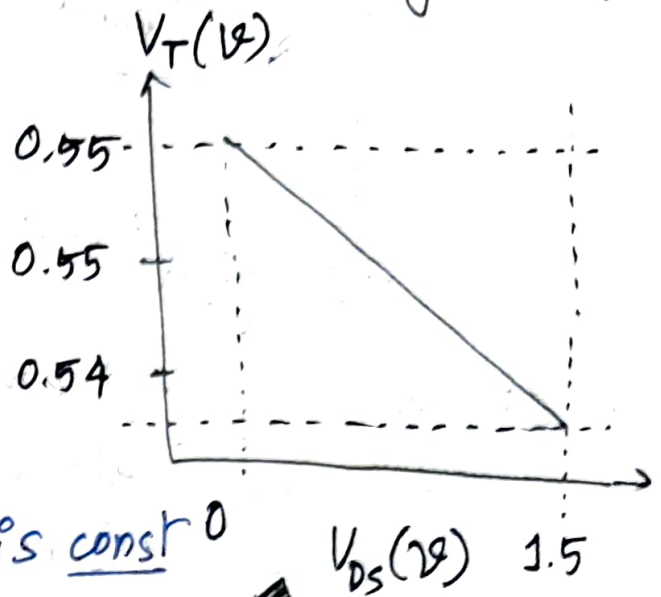
Que

how is there a current below $0.5V$???

(Assumption of $I_D = 0$ if $V_{gs} < V_{th}$ is not valid)

Drain - Induced Barrier Lowering (DIBL) effect :

- Relationship b/w threshold voltage & V_{DS} .



Assumption that threshold voltage is const⁰ is failed

→ it depends on V_{DS}

due to V_{DS} .

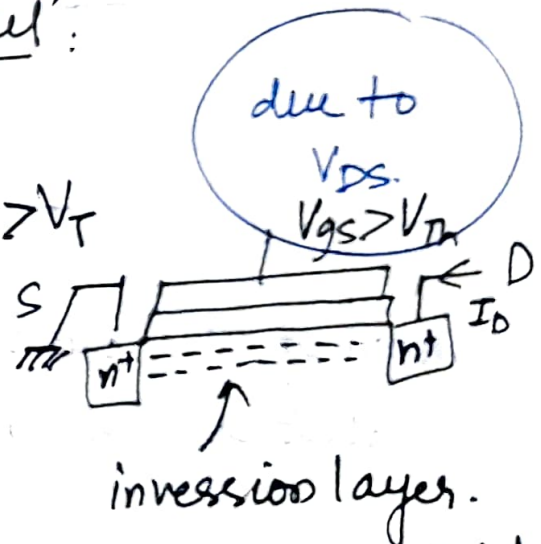
Short-channel MOSFET

* Velocity Saturation Effect:

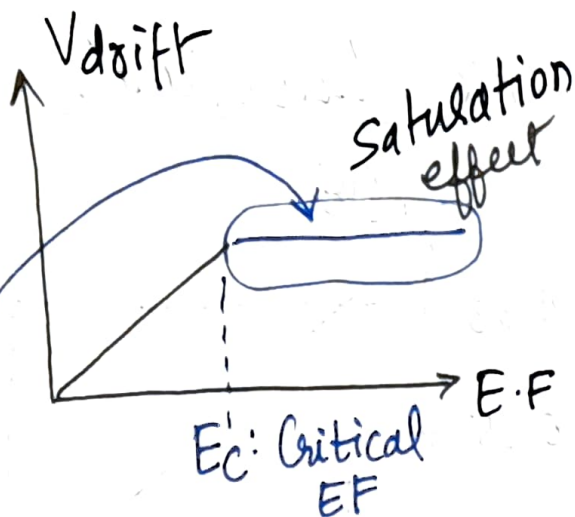
↳ Ca₂ of the V_{DS}

✓ channel is created: $V_{GS} > V_T$

✓ stronger inversion happens. Current flows from D to S.



→ related to drift velocity & E-field



$$\uparrow V_{drift} \propto E \uparrow$$

$$\hookrightarrow I_D = \frac{1}{2} K'_n \left(\frac{W}{L} \right) (V_{GS} - V_{th})^2$$

~~EF is defined as~~

- E is the force present / created b/w two terminals when there is a pot difference.
- Drift velocity is velocity with which e^- move from one terminal to another

$$E_L = \frac{V_{DS}}{L} \quad \text{for small } L \rightarrow$$

a small variation
b/w drain & the source.

— creates a large $E \cdot F$

- ✓ mass of the electron becomes large
(that creates current)

velocity of the
carrier gets
saturated

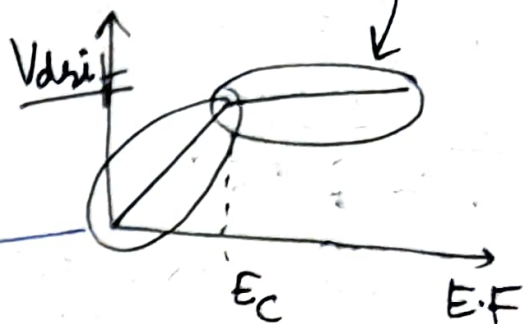
- Critical E (E_c) $\hookrightarrow$ point beyond which if you try to have large E we get velocity of carrier becomes saturated.

(i)

$E_L = \frac{V_{ds}}{L(\text{small})} \rightarrow \text{causes } I_D \propto (V_{gs} - V_{th})$

Before : L is normal

$E_F < E_C$



(ii)

$I_D \propto (V_{gs} - V_{th})^2$

Conclusion

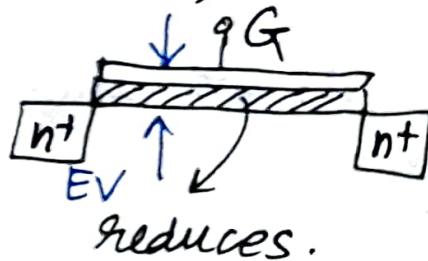
- I_D has linear dep on V_{gs}
- I_D has (or weak) no dep on L

Mobility degradation

due to V_{gs} ✓

• in newer technologies (eg., 45nm, 28nm, 2nm etc..) thickness of oxide reduces

E_v : Vertical electric field b/w oxide & layer.

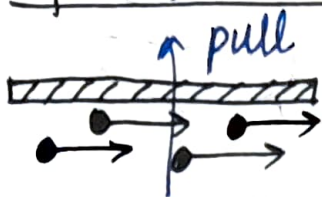


$E_v = \frac{V_{gs}}{t_{ox}}$ → thickness of oxide.

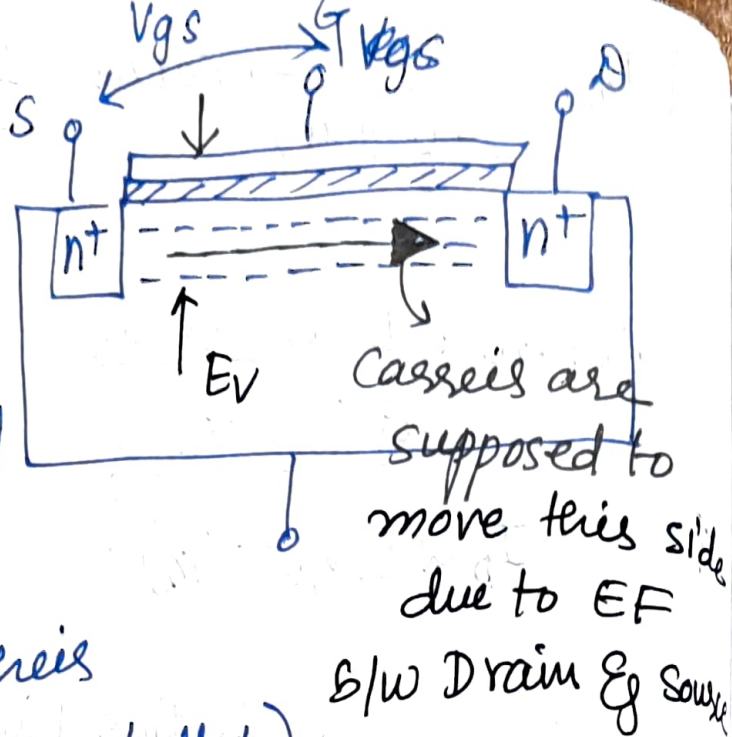
* if thickness reduces → $E \cdot F$ increases!

* ~~higher current flow.~~

if E_v increases :



carriers are forced to move vertical (due to attraction) that degrades their mobility (moving from left to right)

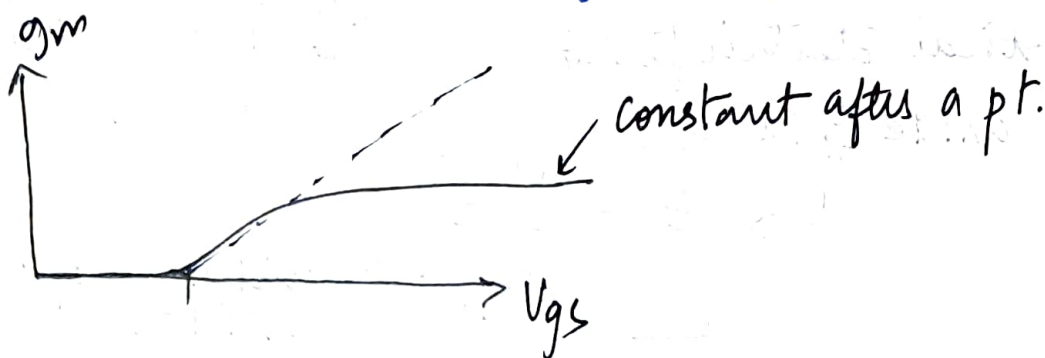


g_m saturates at larger V_{ov}

→ Linear Rather than Quadratic characteristic

whit; acc to sq. law $g_m \propto V_{ov}$

— as we increase $V_{gs} > V_T$ ($V_{gs} - V_{Th}$)



∴ $I_d \propto V_{gs}$ & derivation of I_d w.r.t V_{gs} (gives us g_m) gives constant.

✓

MOSFET g_m/I_D

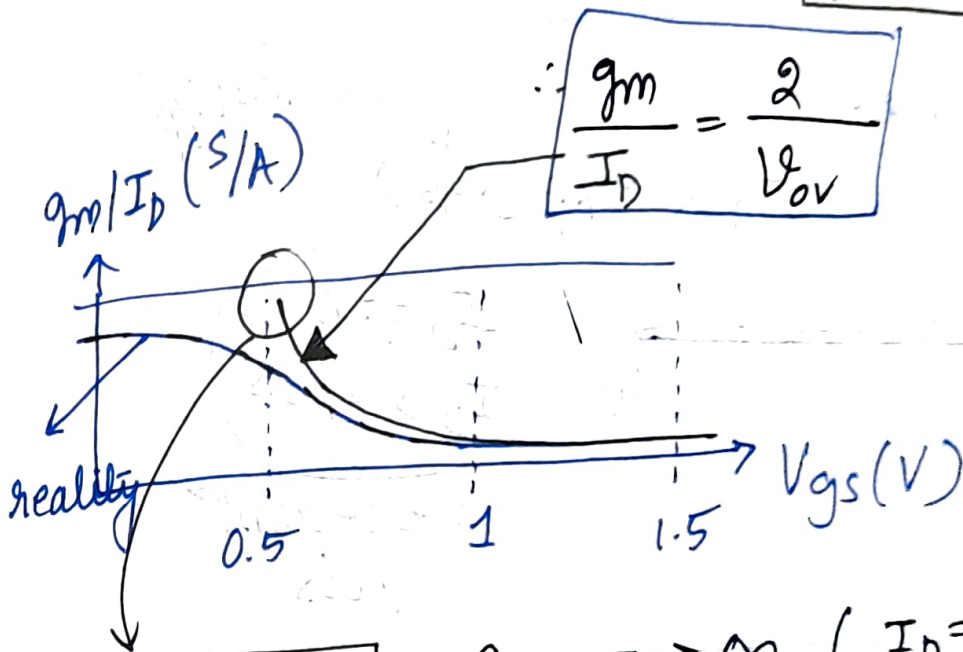
* square law fails in 'strong inversion' region
↳ * in shorter lengths.

Square Law

$$I_D = \frac{1}{2} k'_n \cdot \left(\frac{W}{L}\right) \cdot V_{ov}^2$$

$$g_m = \frac{\partial I_D}{\partial V_{gs}} = k'_n \left(\frac{W}{L}\right) V_{ov}$$

$$g_m \propto V_{ov}$$



• at $V_{gs} = V_{Th}$: $g_m/I_D \rightarrow \infty$ ($I_D = 0$)

• in reality $\rightarrow$ it doesn't tend to ∞ .

• at $V_{gs} < V_{Th}$ $\rightarrow g_m/I_D = \text{constant}$ (graph)

TE $\propto$ Slope of I_C on log scale

$$g_m/I_C \propto \frac{\partial \log_b(I_C)}{\partial V_{BE}} \quad (\text{BJT})$$

• slope = constant

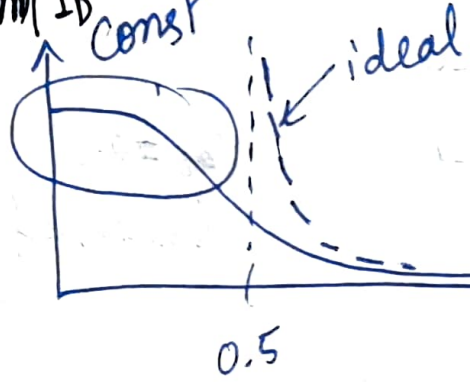
$$\frac{g_m}{I_C} = \frac{1}{U_T}$$

control input

$$I_C \propto e^{V_{BE}}$$

If: $g_m/I_C = \text{constant}$;

g_m/I_D const



$$I_D \propto \exp(V_{gs})$$

control i/p

$$I_D \neq 0$$

$$I_D \propto e^{V_{gs}}$$

we learnt this

Conclusion

for $V_{gs} < V_{th}$: weak inversion region
 $I_D \neq 0$

* g_m/I_D depends on bias point:

- Large in weak inversion ($V_{gs} < V_{th}$)
- Small in strong inversion ($V_{gs} > V_{th}$)

$$I_D \propto e^{V_{gs}}$$

$$I_D \propto (V_{gs} - V_{th})^2$$

3rd region is Moderate inversion (MI)
* current flow is defined by diffusion current in (BJT)

* V_{gs} : 0 to V_{DD}

* TE defines $\begin{cases} \text{speed} \\ \text{noise} \\ \text{behaviors of circuit} \end{cases}$

* Range of g_m/I_d which conveys diff types of inversion.

$g_m/I_d \rightarrow \text{small} \rightarrow \text{stronger inv.}$
 $\rightarrow \text{large} \rightarrow \text{weaker inv.}$
 $\rightarrow \text{mid} \rightarrow \text{moderate.}$

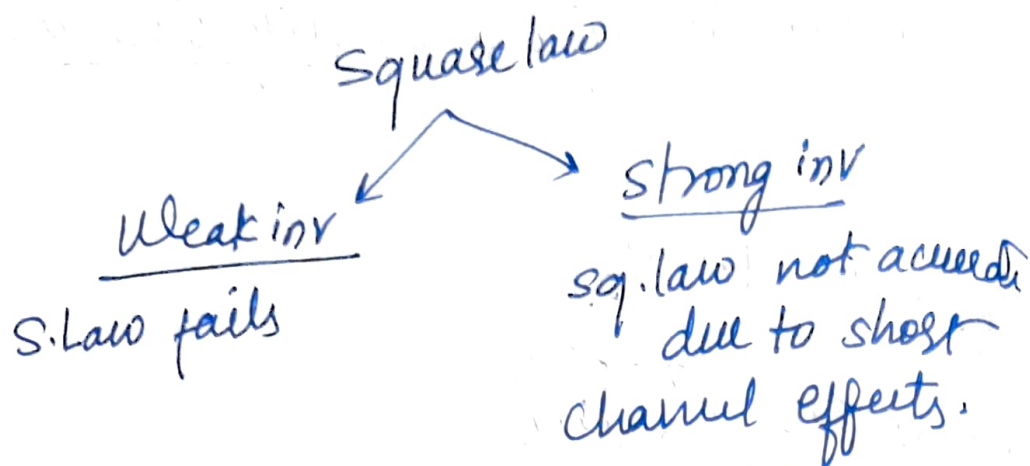
Roughly define the range

10 to 20 $\longrightarrow$ MI

< 10 $\longrightarrow$ SI

> 20 $\longrightarrow$ WI

Square Law End-of-life?



- we use relatively long L & low V_{ov} .
- short channel effects are less pronounced.
- Simple model provides a great deal of intuition that is necessary in Analog design

Square law helps us understand "trends"

$$A_V = g_m R_o$$

$$g_m \uparrow \quad R_o \uparrow$$

$$\begin{aligned} g_m &\propto W \\ &\propto V_{ov} \\ &\propto I_D \end{aligned}$$

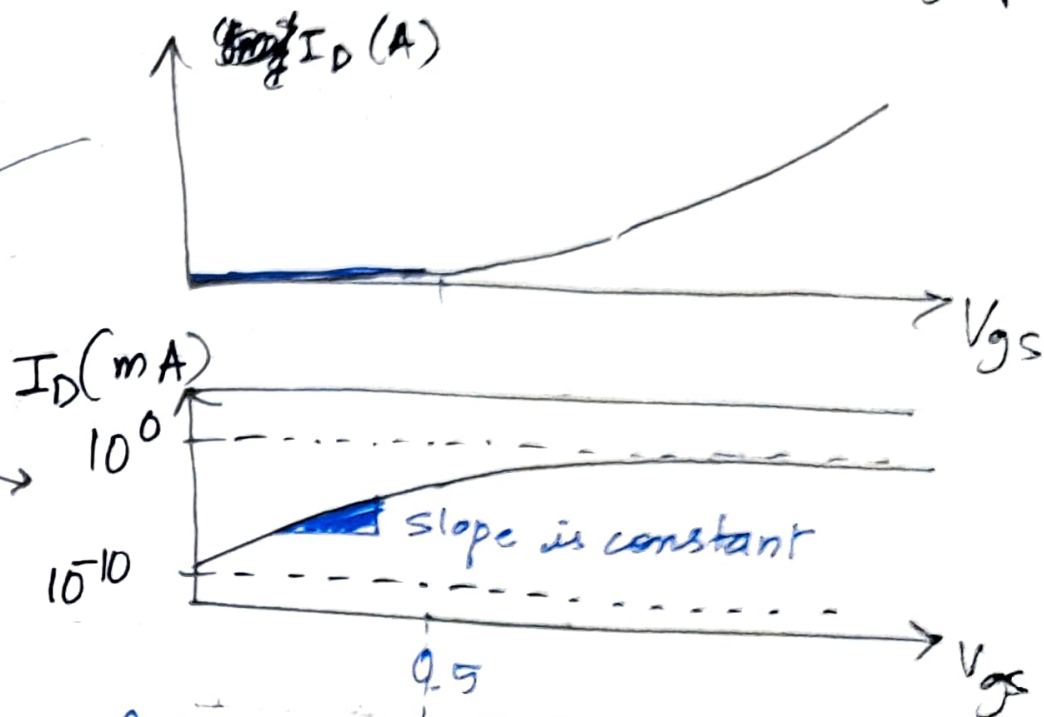
} diff ways to increase g_m to increase gain

* the square law eqⁿ gives this relations

Mosfet inversion Region.

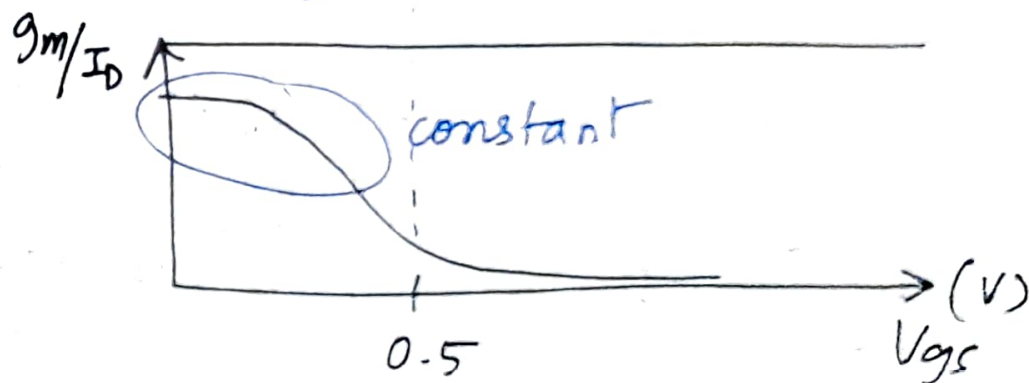
* g_m/I_D saturates in weaker inversion graph.

logarithmic graph.



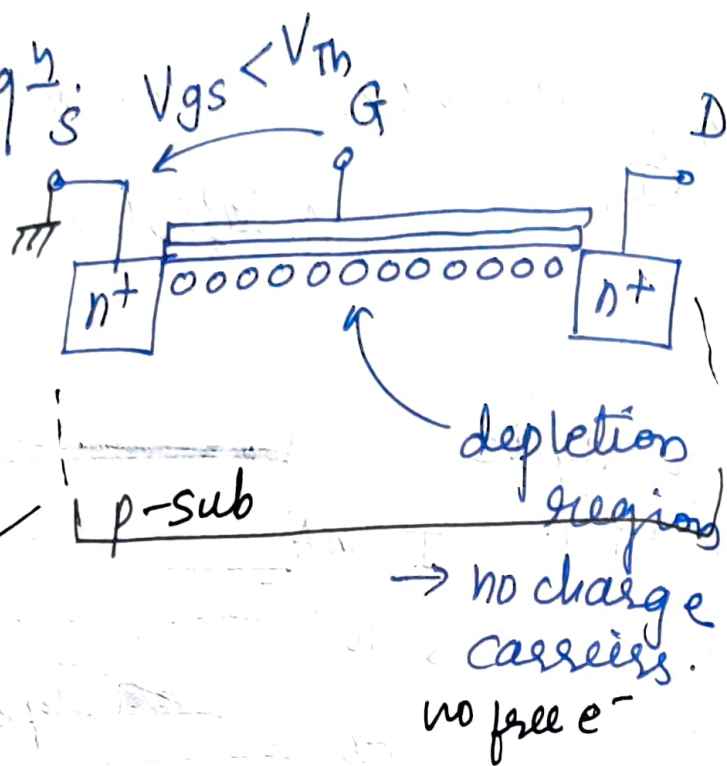
whr

$g_m/I_D \Rightarrow$ slope of log scale of $I_D \Rightarrow$ constant $\ln(I_D)$

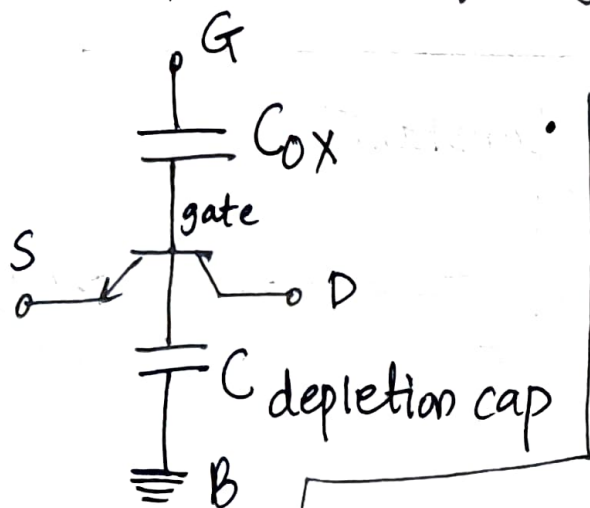


Subthreshold Operation (Weak Inversion)

derive current Eq $\frac{\eta}{S}$; $V_{GS} < V_{Th}$



• Base of BJT is getting connected to gate termi.



$$V_{BE} = V_{GS} \frac{C_{ox}}{C_{ox} + C_{dep}} = \frac{V_{GS}}{\eta}$$

capacitor divides formula

slope factor

$$\eta = \frac{C_{ox} + C_{dep}}{C_{ox}} > 1 \Rightarrow V_{BE} \text{ smaller than } V_{GS}$$

$$I_D \approx I_{off} \exp(V_{BE}/U_T) = I_{off} \exp\left(\frac{V_{GS}}{\eta U_T}\right)$$

$$I_D \approx I_{off} \exp\left(\frac{V_{gs}}{\eta U_T}\right)$$

→ derivative: $g_m = \frac{\partial I_D}{\partial V_{gs}} = \frac{I_D}{\eta U_T}$

$$g_m/I_D = \frac{1}{\eta U_T} = \frac{38.5}{\eta}$$

WOT WI

Gain of MOSFET. { use case of WI }

$$|A_v| = g_m r_o = g_m \cdot \frac{V_A}{I_D}$$

← Early voltage
← Bias current (in WI)

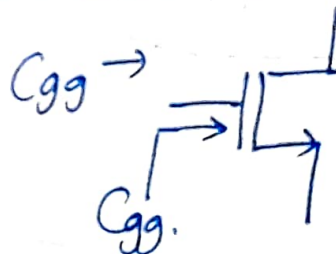
$$A_v \propto g_m/I$$

$$\rightarrow |A_v| = \frac{g_m}{I_D} \cdot V_A = \frac{V_A}{\eta U_T} \quad (\text{in WI})$$

$$\rightarrow |A_v| = \frac{g_m}{I_D} \cdot V_A = \frac{2V_A}{V_{ov}} \quad (\text{in SI}) \quad \left\{ I_D \sim I_{off} \exp\left(\frac{V_{gs}}{\eta U_T}\right) \right.$$

Que. if for higher gain do we operate in WI??

Transient freq $f_T = \frac{g_m}{2\pi C_{gg}}$



$10^0 \log(I_D)$

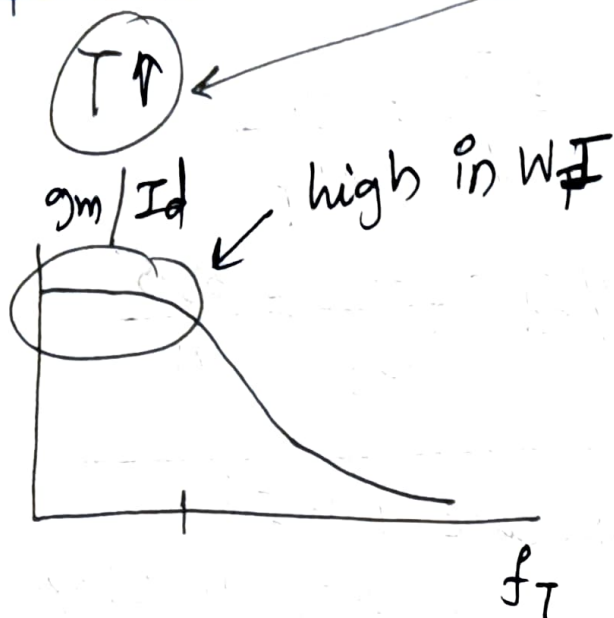
degradation as WI $\rightarrow g_m$ also smaller.

10^{-10}

V_{gs}

• speed also decreases.

$$\downarrow f_T = \frac{g_m \downarrow}{2\pi C_{gg}}$$



Que what if we increase W to increase Id?

→ $I_D \propto W$: but I_D is degrading exponent as in WI

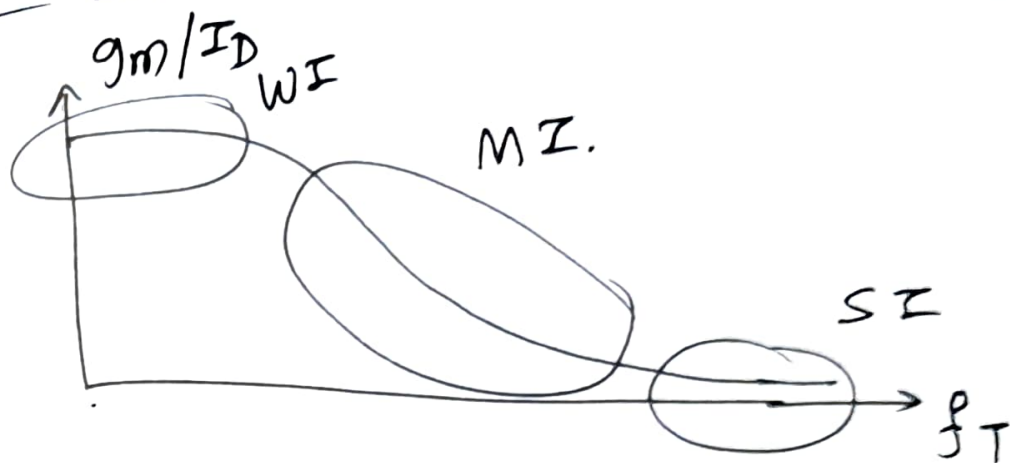
: so we have to increase W exponentially to increase Id

→ to increase speed

$C_{gg} \propto \text{Width}$

* people don't work in WI altho we have higher gm but speed decreases

Examples : In bio-med devices use weakes I.
desires to get very low power
↓
Spend very low I_D



* Current in MI — (MI can't be defined)

Bridge the current I_D from WI & SI

Compact models : EKV, ACM

to create I_D in MI, SI, WI } versatile
few parameters considered only.
(2nd order effects ignored).

Summary

